

Ubuntu-base(20.04 或 22.04)移植



版本历史

版本	版本更新说明	责任人	校审人	发布时间
V1.0	初次	易百纳板卡团队	易百纳板卡团队	2025 年 10 月 29 日

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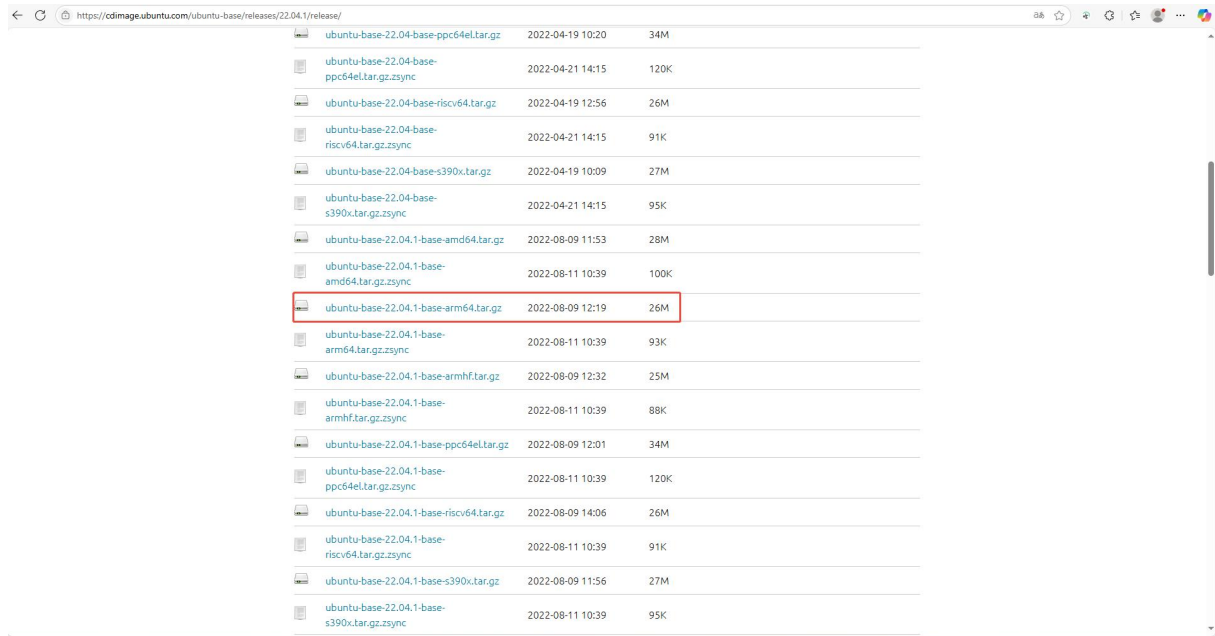
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1. Ubuntu-base 文件下载

官网下载链接: <https://cdimage.ubuntu.com/ubuntu-base/releases/>



ubuntu-base-22.04-base-ppc64el.tar.gz	2022-04-19 10:20	34M
ubuntu-base-22.04-base-ppc64el.tar.gz.zsync	2022-04-21 14:15	120K
ubuntu-base-22.04-base-riscv64.tar.gz	2022-04-19 12:56	26M
ubuntu-base-22.04-base-riscv64.tar.gz.zsync	2022-04-21 14:15	91K
ubuntu-base-22.04-base-s390x.tar.gz	2022-04-19 10:09	27M
ubuntu-base-22.04-base-s390x.tar.gz.zsync	2022-04-21 14:15	95K
ubuntu-base-22.04.1-base-amd64.tar.gz	2022-08-09 11:53	28M
ubuntu-base-22.04.1-base-amd64.tar.gz.zsync	2022-08-11 10:39	100K
ubuntu-base-22.04.1-base-arm64.tar.gz	2022-08-09 12:19	26M
ubuntu-base-22.04.1-base-arm64.tar.gz.zsync	2022-08-11 10:39	93K
ubuntu-base-22.04.1-base-armhf.tar.gz	2022-08-09 12:32	25M
ubuntu-base-22.04.1-base-armhf.tar.gz.zsync	2022-08-11 10:39	88K
ubuntu-base-22.04.1-base-ppc64el.tar.gz	2022-08-09 12:01	34M
ubuntu-base-22.04.1-base-ppc64el.tar.gz.zsync	2022-08-11 10:39	120K
ubuntu-base-22.04.1-base-riscv64.tar.gz	2022-08-09 14:06	26M
ubuntu-base-22.04.1-base-riscv64.tar.gz.zsync	2022-08-11 10:39	91K
ubuntu-base-22.04.1-base-s390x.tar.gz	2022-08-09 11:56	27M
ubuntu-base-22.04.1-base-s390x.tar.gz.zsync	2022-08-11 10:39	95K

下载 ubuntu-base-22.04.1-base-arm64.tar.gz 并解压至 ubuntu2204

```
sunshine@sunshine-vm:~/Workspace/Sunshine/Ubuntu$ ls
ubuntu-base-22.04.1-base-arm64.tar.gz
sunshine@sunshine-vm:~/Workspace/Sunshine/Ubuntu$ mkdir ubuntu2204
sunshine@sunshine-vm:~/Workspace/Sunshine/Ubuntu$ ls
ubuntu2204  ubuntu-base-22.04.1-base-arm64.tar.gz
sunshine@sunshine-vm:~/Workspace/Sunshine/Ubuntu$ tar -xvf ubuntu-base-22.04.1-base-arm64.tar.gz -C ubuntu2204/
```

2. Ubuntu 系统环境搭建

2.1. 安装 qemu 模拟器:

```
sudo apt update
sudo apt install qemu-user-static binfmt-support
```

```
sunshine@sunshine-vm:~/Workspace/Sunshine/Ubuntu$ sudo apt install qemu-user-static binfmt-support
[sudo] sunshine 的密码:
正在读取软件包列表... 完成
正在分析软件包的依赖关系树
正在读取状态信息... 完成
下列【新】软件包将被安装:
  binfmt-support qemu-user-static
升级了 0 个软件包，新安装了 2 个软件包，要卸载 0 个软件包，有 3 个软件包未被升级。
需要下载 21.4 MB 的归档。
解压后会消耗 208 MB 的额外空间。
获取:1 https://mirrors.tuna.tsinghua.edu.cn/ubuntu focal/universe amd64 binfmt-support amd64 2.2.0-2 [58.2 kB]
获取:2 https://mirrors.tuna.tsinghua.edu.cn/ubuntu focal-updates/universe amd64 qemu-user-static amd64 1:4.2-3ubuntu6.30 [21.3 MB]
已下载 21.4 MB，耗时 6秒 (3,698 kB/s)
正在选中未选择的软件包 binfmt-support。
(正在读取数据库 ... 系统当前共安装有 223928 个文件和目录。)
准备解压 .../binfmt-support_2.2.0-2_amd64.deb ...
正在解压 binfmt-support (2.2.0-2) ...
正在选中未选择的软件包 qemu-user-static。
准备解压 .../qemu-user-static_1%3a4.2-3ubuntu6.30_amd64.deb ...
正在解压 qemu-user-static (1:4.2-3ubuntu6.30) ...
正在设置 qemu-user-static (1:4.2-3ubuntu6.30) ...
正在设置 binfmt-support (2.2.0-2) ...
Created symlink /etc/systemd/system/multi-user.target.wants/binfmt-support.service → /lib/systemd/system/binfmt-support.service.
正在处理用于 man-db (2.9.1-1) 的触发器 ...
正在处理用于 systemd (245.4-4ubuntu3.24) 的触发器 ...
```

将 qemu 可执行文件复制进文件系统:

```
sudo cp /usr/bin/qemu-aarch64-static ubuntu2204/usr/bin/
```

```
sunshine@sunshine-vm:~/Workspace/Sunshine/Ubuntu$ which qemu-aarch64-static
/usr/bin/qemu-aarch64-static
sunshine@sunshine-vm:~/Workspace/Sunshine/Ubuntu$ sudo cp /usr/bin/qemu-aarch64-static ubuntu2204/usr/bin/
[sudo] sunshine 的密码:
sunshine@sunshine-vm:~/Workspace/Sunshine/Ubuntu$
```

2.2. 换源并配置 DNS

1). 换源

```
cp ubuntu2204/etc/apt/sources.list
ubuntu2204/etc/apt/sources.ubuntu
vi ubuntu2204/etc/apt/sources.list
```

查看 Ubuntu 开发代号，替换 sources.list 的内容，例如 Ubuntu20.04 需要将 'jammy' 改成 'focal'。

```
sunshine@sunshine-vm:~/Workspace/Sunshine/Ubuntu$ cat Ubuntu20.04/etc/os-release
NAME="Ubuntu"
VERSION="20.04.1 LTS (Focal Fossa)"
ID=ubuntu
ID_LIKE=debian
PRETTY_NAME="Ubuntu 20.04.1 LTS"
VERSION_ID="20.04"
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
VERSION_CODENAME=focal
UBUNTU_CODENAME=focal
sunshine@sunshine-vm:~/Workspace/Sunshine/Ubuntu$ cat Ubuntu22.04/etc/os-release
NAME="Ubuntu"
VERSION="22.04"
VERSION_ID="22.04"
VERSION_CODENAME=jammy
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=jammy
sunshine@sunshine-vm:~/Workspace/Sunshine/Ubuntu$ cat Ubuntu24.04/etc/os-release
NAME="Ubuntu"
VERSION="24.04"
VERSION_ID="24.04"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
```

```
# 默认注释了源码镜像以提高 apt update 速度，如有需要可自行取消注释

deb http://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports/ jammy main restricted universe
multiverse

# deb-src http://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports/ jammy main restricted
universe multiverse

deb http://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports/ jammy-updates main restricted
universe multiverse

# deb-src http://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports/ jammy-updates main
restricted universe multiverse

deb http://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports/ jammy-backports main restricted
universe multiverse

# deb-src http://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports/ jammy-backports main
restricted universe multiverse

deb http://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports/ jammy-security main restricted
universe multiverse

# deb-src http://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports/ jammy-security main
restricted universe multiverse

# 预发布软件源，不建议启用

# deb http://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports/ jammy-proposed main restricted
universe multiverse

# # deb-src http://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports/ jammy-proposed main
restricted universe multiverse
```

```
deb http://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports/ jammy main restricted universe multiverse
# deb-src http://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports/ jammy main restricted universe multiverse
deb http://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports/ jammy-updates main restricted universe multiverse
# deb-src http://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports/ jammy-updates main restricted universe multiverse
deb http://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports/ jammy-backports main restricted universe multiverse
# deb-src http://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports/ jammy-backports main restricted universe multiverse

deb http://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports/ jammy-security main restricted universe multiverse
# deb-src http://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports/ jammy-security main restricted universe multiverse

# deb http://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports/ jammy-proposed main restricted universe multiverse
# # deb-src http://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports/ jammy-proposed main restricted universe multiverse
~
```


2). 添加 dns 域名解析服务器地址

```
nameserver 114.114.114.114  
  
nameserver 8.8.8.8
```

```
sunshine@sunshine-vm:~/Workspace/Sunshine/Ubuntu$ vi ubuntu2204/etc/resolv.conf  
sunshine@sunshine-vm:~/Workspace/Sunshine/Ubuntu$ cat ubuntu2204/etc/resolv.conf  
nameserver 114.114.114.114  
nameserver 8.8.8.8  
sunshine@sunshine-vm:~/Workspace/Sunshine/Ubuntu$
```

3. 在宿主机里挂载 Ubuntu 文件系统

3.1 新建 ubuntu_chroot.sh


```
#!/bin/bash
mnt() {
    echo "MOUNTING"
    mount -t proc /proc ${2}/proc
    mount -t sysfs /sys ${2}/sys
    mount -o bind /dev ${2}/dev
    mount -o bind /dev/pts ${2}/dev/pts
    chroot ${2}
}
umnt() {
    echo "UNMOUNTING"
    umount ${2}/proc
    umount ${2}/sys
    umount ${2}/dev/pts
    umount ${2}/dev
}
if [ "$1" = "-m" ] && [ -n "$2" ]; then
    mnt $1 $2
    echo "mnt -m pwd"
elif [ "$1" = "-u" ] && [ -n "$2" ]; then
    umnt $1 $2
    echo "umnt -u pwd"
else
    echo ""
    echo "Either 1st, 2nd or both parameters"
    echo ""
    echo "1st parameter can be one of these: -m"
    echo "2nd parameter is the full path of root"
    echo ""
    echo "For example: chroot_ubuntu -m ubuntu22.04"
    echo ""
    echo "1st parameter: ${1}"
    echo "2nd parameter: ${2}"
fi
```

```
./ubuntu_chroot.sh -m ubuntu2004 # 运行命令挂载，进入 ubuntu2004 系统。
./ubuntu_chroot.sh -u ubuntu2004 # exit 退出后运行命令卸载！ 非常重要，退出后及时卸载
```

```
sunshine@sunshine-vm:~/WorkSpace/Sunshine/Ubuntu$ ls
ubuntu2204  ubuntu-base-22.04.1-base-arm64.tar.gz  ubuntu_chroot.sh
sunshine@sunshine-vm:~/WorkSpace/Sunshine/Ubuntu$ sudo ./ubuntu_chroot.sh -m ubuntu2204/
MOUNTING
root@sunshine-vm:/# exit
exit
mnt -m pwd
sunshine@sunshine-vm:~/WorkSpace/Sunshine/Ubuntu$ sudo ./ubuntu_chroot.sh -u ubuntu2204/
UNMOUNTING
umnt -u pwd
sunshine@sunshine-vm:~/WorkSpace/Sunshine/Ubuntu$ sudo ./ubuntu_chroot.sh -m ubuntu2204/
MOUNTING
root@sunshine-vm:/#
```

3.2 apt 安装基础应用

3.2.1 更新软件包

```
apt-get update
apt upgrade -y
```

可能会有报错：

```
root@sunshine-vm:/# apt-get update
Get:1 http://ports.ubuntu.com/ubuntu-ports jammy InRelease [270 kB]
Err:1 http://ports.ubuntu.com/ubuntu-ports jammy InRelease
  Couldn't create temporary file /tmp/apt.conf.XtUgcT for passing config to apt-key
Get:2 http://ports.ubuntu.com/ubuntu-ports jammy-updates InRelease [128 kB]
Err:2 http://ports.ubuntu.com/ubuntu-ports jammy-updates InRelease
  Couldn't create temporary file /tmp/apt.conf.Dbq4Fu for passing config to apt-key
Get:3 http://ports.ubuntu.com/ubuntu-ports jammy-backports InRelease [127 kB]
Err:3 http://ports.ubuntu.com/ubuntu-ports jammy-backports InRelease
  Couldn't create temporary file /tmp/apt.conf.wRaaI3 for passing config to apt-key
Get:4 http://ports.ubuntu.com/ubuntu-ports jammy-security InRelease [129 kB]
Err:4 http://ports.ubuntu.com/ubuntu-ports jammy-security InRelease
  Couldn't create temporary file /tmp/apt.conf.6GKo4K for passing config to apt-key
Reading package lists... Done
W: GPG error: http://ports.ubuntu.com/ubuntu-ports jammy InRelease: Couldn't create temporary file /tmp/apt.conf.XtUgcT for passing config to apt-key
E: The repository 'http://ports.ubuntu.com/ubuntu-ports jammy InRelease' is not signed.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
W: GPG error: http://ports.ubuntu.com/ubuntu-ports jammy-updates InRelease: Couldn't create temporary file /tmp/apt.conf.Dbq4Fu for passing config to apt-key
E: The repository 'http://ports.ubuntu.com/ubuntu-ports jammy-updates InRelease' is not signed.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
W: GPG error: http://ports.ubuntu.com/ubuntu-ports jammy-backports InRelease: Couldn't create temporary file /tmp/apt.conf.wRaaI3 for passing config to apt-key
E: The repository 'http://ports.ubuntu.com/ubuntu-ports jammy-backports InRelease' is not signed.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
W: GPG error: http://ports.ubuntu.com/ubuntu-ports jammy-security InRelease: Couldn't create temporary file /tmp/apt.conf.6GKo4K for passing config to apt-key
E: The repository 'http://ports.ubuntu.com/ubuntu-ports jammy-security InRelease' is not signed.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
```

解决方式：

```
chown root:root /tmp
chmod 1777 /tmp
```

3.2.2 安装一些基础组件

```
apt-get install vim
apt-get install net-tools
apt-get install iputils-ping

apt-get install systemd
apt-get install udev
apt-get install haveged
apt-get install kmod      end

apt-get install telnetd
apt-get install ssh
apt-get install udhcpd
```

(1) 安装 systemd 之后，若没有/sbin/init 文件，需创建启动程序链接。

```
root@sunshine-vm:/# ls /sbin/init -lh
lrwxrwxrwx 1 root root 20 Jun  4 14:17 /sbin/init -> /lib/systemd/systemd
root@sunshine-vm:/#
```

```
ln -s /lib/systemd/systemd /sbin/init
```

(2) 设置登录密码

串口/telnet/ssh 连接时需要使用用户密码登录

```
root@sunshine-vm:/# passwd root
New password:
Retype new password:
passwd: password updated successfully
root@sunshine-vm:/#
```

(3) telnet 连接需要修改/usr/lib/telnetlogin 的权限

```
root@sunshine-vm:/# ls -lh /usr/lib/telnetlogin
-rwsr-xr-- 1 root telnetd 11K Mar 24 2022 /usr/lib/telnetlogin
root@sunshine-vm:/# chmod 4755 /usr/lib/telnetlogin
root@sunshine-vm:/# ls -lh /usr/lib/telnetlogin
-rwsr-xr-x 1 root telnetd 11K Mar 24 2022 /usr/lib/telnetlogin
root@sunshine-vm:/#
```

ssh 连接需要修改/etc/ssh/sshd_config 中 PermitRootLogin 部分

```
# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin prohibit-password
PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
```

(4) 替换 dash 为 bash: 解决 sh 脚本执行问题

```
rm /bin/sh
ln -s bash /bin/sh
```

```
root@sunshine-vm:/# ls -lh /bin/sh
lrwxrwxrwx 1 1000 1000 4 Mar 23 2022 /bin/sh -> dash
root@sunshine-vm:/# rm /bin/sh
root@sunshine-vm:/# ln -s bash /bin/sh
root@sunshine-vm:/# ls -lh /bin/sh
lrwxrwxrwx 1 root root 4 Jul 24 09:05 /bin/sh -> bash
root@sunshine-vm:/#
```

(5) 添加启动脚本/etc/rc.local, 并添加可执行权限

```
#!/bin/bash

ifconfig eth0 up
ifconfig lo up

ifconfig eth0 192.168.5.100
```



```
root@sunshine-vm:/# vi /etc/rc.local
root@sunshine-vm:/# cat /etc/rc.local
#!/bin/bash

ifconfig eth0 up
ifconfig lo up

ifconfig eth0 192.168.5.100

root@sunshine-vm:/# chmod +x /etc/rc.local
root@sunshine-vm:/#
```

4. 快速打包验证

4.1 检查文件系统大小

```
sunshine@sunshine-vm:~/WorkSpace/Sunshine/Ubuntu$ sudo du -sh *
711M    ubuntu2204
27M     ubuntu-base-22.04.1-base-arm64.tar.gz
4.0K    ubuntu_chroot.sh
sunshine@sunshine-vm:~/WorkSpace/Sunshine/Ubuntu$
```

4.2 打包文件系统

```
#打包 ext4 格式
dd if=/dev/zero of=./ubuntu_2204_rootfs_1G.ext4 bs=1M count=1024
fakeroot mkfs.ext4 -d ubuntu2204 ubuntu_2204_rootfs_1G.ext4
```

fakeroot 会让 mkfs.ext4 以“看起来是 root”的权限运行，但实际不会影响你的主机文件归属，文件再板子上归属为 root。

```
sunshine@sunshine-vm:~/WorkSpace/Sunshine/Ubuntu$ dd if=/dev/zero of=./ubuntu_2204_rootfs_1G.ext4 bs=1M count=1024
记录了1024+0 的读入
记录了1024+0 的写出
1073741824字节 (1.1 GB, 1.0 GiB) 已复制, 0.873844 s, 1.2 GB/s
sunshine@sunshine-vm:~/WorkSpace/Sunshine/Ubuntu$
sunshine@sunshine-vm:~/WorkSpace/Sunshine/Ubuntu$ fakeroot mkfs.ext4 -d ubuntu2204 ubuntu_2204_rootfs_1G.ext4
mke2fs 1.45.5 (07-Jan-2020)
丢弃设备块: 完成
创建含有 262144 个块 (每块 4k) 和 65536 个 inode 的文件系统
文件系统 UUID: 1c94ebe0-f641-47ff-b208-21aaf94e7836
超级块的备份存储于下列块:
    32768, 98304, 163840, 229376

正在分配组表: 完成
正在写入 inode表: 完成
创建日志 (8192 个块): 完成
将文件复制到设备: 完成
写入超级块和文件系统账户统计信息: 已完成
sunshine@sunshine-vm:~/WorkSpace/Sunshine/Ubuntu$
```

4.3 烧录验证文件系统

(1) 烧录完成检查是否能正常启动，有无报错，启动需要内核开启 CGROUP，参考第五节

```
[ OK ] Reached target System Initialization.
[ OK ] Started Daily Cleanup of Temporary Directories.
[ OK ] Reached target Basic System.
[ OK ] Reached target System Time Set.
[ OK ] Started Daily apt download activities.
[ OK ] Started Daily apt upgrade and clean activities.
[ OK ] Started Daily dpkg database backup timer.
[ OK ] Started Periodic ext4 Onli_data Check for All Filesystems.
[ OK ] Started Discard unused blocks once a week.
[ OK ] Started Message of the Day.
[ OK ] Reached target Timer Units.
[ OK ] Listening on D-Bus System Message Bus Socket.
[ OK ] Started D-Bus System Message Bus.
Starting Remove Stale Onli_ext4 Metadata Check Snapshots...
Starting Internet superserver...
Starting Dispatcher daemon for systemd-networkd...
Starting /etc/rc.local Compatibility...
Starting OpenBSD Secure Shell server...
Starting User Login Management...
Starting Permit User Sessions...
[ OK ] Started Network Name Resolution.
[ OK ] Started Internet superserver.
[ OK ] Finished Load/Save Random Seed.
[ OK ] Finished Remove Stale Onli_ext4 Metadata Check Snapshots.
[ OK ] Started /etc/rc.local Compatibility.
[ OK ] Finished Permit User Sessions.
[ OK ] Reached target Host and Network Name Lookups.
[ OK ] Started Getty on tty1.
[ OK ] Started Serial Getty on ttyAMA0.
[ OK ] Reached target Login Prompts.
[ OK ] Started OpenBSD Secure Shell server.
[ OK ] Started User Login Management.
[ OK ] Started Dispatcher daemon for systemd-networkd.
[ OK ] Reached target Multi-User System.
[ OK ] Reached target Graphical Interface.
Starting Record Runlevel Change in UTMP...
[ OK ] Finished Record Runlevel Change in UTMP.

Ubuntu 22.04.1 LTS localhost.localdomain ttyAMA0
localhost login: █
```

注意： Ubuntu20.04 会卡在 A start job is running for Load/Save Random Seed 。

```
[ OK ] Finished Update UTMP about System Boot/Shutdown.
[ OK ] Found device /dev/ttyAMA0.
[ OK ] Listening on Load/Save RF Watch Status /dev/rfkill Watch.
[ OK ] Started Network Time Synchronization.
[ OK ] Reached target System Time Set.
[ OK ] Reached target System Time Synchronized.
[ OK ] Started Network Name Resolution.
[ OK ] Reached target Host and Network Name Lookups.
[ *** ] A start job is running for Load/Save Random Seed (1min 14s / 10min 1s)
█
```

解决方式：考虑修改 haveged.service 与 systemd-random-seed.service 的启动顺序与依赖关系。
修改启动顺序为 haveged 在 random 之前。

```
vi /lib/systemd/system/systemd-random-seed.service
vi /lib/systemd/system/haveged.service
```

```
# SPDX-License-Identifier: LGPL-2.1+
#
# This file is part of systemd.
#
# systemd is free software; you can redistribute it and/or modify it
# under the terms of the GNU Lesser General Public License as published by
# the Free Software Foundation; either version 2.1 of the License, or
# (at your option) any later version.

[Unit]
Description=Load/Save Random Seed
Documentation=man:systemd-random-seed.service(8) man:random(4)
DefaultDependencies=no
RequiresMountsFor=/var/lib/systemd/random-seed
Conflicts=shutdown.target
After=systemd-remount-fs.service haveged.service
Before=shutdown.target
ConditionVirtualization=!container

[Service]
Type=oneshot
RemainAfterExit=yes
ExecStart=/lib/systemd/systemd-random-seed load
ExecStop=/lib/systemd/systemd-random-seed save

# This service waits until the kernel's entropy pool is initialized, and may be
# used as ordering barrier for service that require an initialized entropy
# pool. Since initialization can take a while on entropy-starved systems, let's
# increase the time-out substantially here.
TimeoutSec=10min

~

[Unit]
Description=Entropy daemon using the HAVEGE algorithm
Documentation=man:haveged(8) http://www.issihosts.com/haveged/
DefaultDependencies=no
ConditionVirtualization=!container
After=apparmor.service
Before=sysinit.target shutdown.target
Conflicts=shutdown.target

[Service]
EnvironmentFile=-/etc/default/haveged
ExecStart=/usr/sbin/haveged --Foreground --verbose=1 $DAEMON_ARGS
SuccessExitStatus=143
SecureBits=noroot-locked
NoNewPrivileges=yes
CapabilityBoundingSet=CAP_SYS_ADMIN
PrivateTmp=yes
PrivateDevices=yes
PrivateNetwork=yes
ProtectSystem=full
ProtectHome=yes

[Install]
WantedBy=default.target

~
```

(2) 测试 telnetd/ssh


```
Serial-COM9 192.168.5.100 x
Ubuntu 22.04.1 LTS
localhost.localdomain login: root
Password:
Welcome to Ubuntu 22.04.1 LTS (GNU/Linux 4.19.90 aarch64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
Last login: Wed Jun  4 14:17:58 UTC 2025 from 192.168.5.4 on pts/0
root@localhost:~#
```

```
Serial-COM9 192.168.5.100 SSH x
Welcome to Ubuntu 22.04.1 LTS (GNU/Linux 4.19.90 aarch64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
Last login: Wed Jun  4 14:17:51 2025
root@localhost:~#
```

5. 内核开启 CGROUP

内核需要开启 CGROUP，重新编译、烧录。

Search (cgroups)

```
Symbol: CGROUPS [=n]
Type : bool
Prompt: Control Group support
Location:
(1) -> General setup
Defined at init/Kconfig:673
Selects: KERNFS [=y]
Selected by [n]:
- SCHED_AUTOGROUP [=n]
```

```
*** Compiler: aarch64-mix210-linux-gcc (HC&C V1R3C00SPC200B042_20221123) 7.3.0 ***
```

General setup --->

```
*- Patch physical to virtual translations at runtime
System Type --->
Bus support --->
Kernel Features --->
```

< > Kernel .config support

```
(14) Kernel log buffer size (16 => 64KB, 17 => 128KB)
(12) CPU kernel log buffer size contribution (13 => 8 KB, 17 => 128KB)
(13) Temporary per-CPU printk log buffer size (12 => 4KB, 13 => 8KB)
```

[*] Control Group support --->

```
[*] Namespaces support --->
[ ] Checkpoint/restore support
[ ] Automatic process group scheduling
```

6. 项目打包验证

6.1 打包可能遇到的问题

```
sunshine@sunshine-vm:~/Workspace/HiEuler_PI_Firmware_Building$ ./Generate_Image.sh -g
generate a new image
Step: init image path ...
mkdir /home/sunshine/Workspace/HiEuler_PI_Firmware_Building/out/Eulerpi_4GB_Ubuntu_IMAGE_20250722103604 ...
Step: copy logo file ...
Step: copy uboot file ...
Step: generate uboot env ...
Generate /home/sunshine/Workspace/HiEuler_PI_Firmware_Building/out/Eulerpi_4GB_Ubuntu_IMAGE_20250722103604/uboot_env.bin
Step: copy kernel file ...
Step: clean rootfs ...
Step: decompression rootfs ...
Step: copy boardtools file ...
Step: patch overlay file ...
Step: patch opt file ...
Step: emmc filesystem ...
Step: generate ext4fs debugfs ...
记录了1024+0 的写入
记录了1024+0 的写入
1073741824字节 (1.1 GB, 1.0 GiB) 已复制, 7.01487 s, 153 MB/s
mk2fs 1.45.5 (07-Jan-2020)
丢弃设备块: 完成
创建含有 262144 个块 (每块 4k) 和 65536 个 inode 的文件系统
文件系统 UUID: f7ad6a9b-6467-4baa-aced-cb2bf978b6f
超级块的备份存储在下列块:
32768, 98304, 163840, 229376
正在分配组表: 完成
正在写入 inode表: 完成
创建日志 (8192 个块): 完成
写入超级块和文件系统账户统计信息: 已完成
debugfs 1.45.5 (07-Jan-2020)
stat: cannot stat '/home/sunshine/Workspace/HiEuler_PI_Firmware_Building/rootfs/rootfs/usr/lib/systemd/system/systemdxdcryptsetup.slice': No such file or directory
do_write_internal: No such file or directory while opening "/home/sunshine/Workspace/HiEuler_PI_Firmware_Building/rootfs/rootfs/usr/lib/systemd/system/systemdxdcryptsetup.slice" to copy
write: No such file or directory
system-systemdxdcryptsetup.slice: File not found by ext2_lookup
system-systemdxdcryptsetup.slice: File not found by ext2_lookup
system-systemdxdcryptsetup.slice: File not found by ext2_lookup
sunshine-vm:~/Workspace/HiEuler_PI_Firmware_Building$
```

解决方式： 将 x2d 改成 -

```
root@sunshine-vm:/lib/systemd/system# ls system-systemd\x2dcryptsetup.slice
'system-systemd\x2dcryptsetup.slice'
root@sunshine-vm:/lib/systemd/system# mv system-systemd\x2dcryptsetup.slice system-systemd-cryptsetup.slice
root@sunshine-vm:/lib/systemd/system#
```

6.2 网络没有生效

打包/etc/systemd/network/20-static-eth0.network，但是网络没有生效。

```
ln -s /lib/systemd/system/systemd-networkd.service
/etc/systemd/system/multi-user.target.wants/systemd-networkd.service
```

```
root@sunshine-vm:~# ls
root@sunshine-vm:~# ln -s /lib/systemd/system/systemd-networkd.service /etc/systemd/system/multi-user.target.wants/systemd-networkd.service
root@sunshine-vm:~#
```

6.3 telnetd 没有生效

板端修改/usr/lib/telnetlogin 的权限，或者打包脚本里添加。

```
root@localhost:/bin# ls /usr/lib/telnetlogin -lh
-rwsr-xr-x 1 1000 1000 11K Jul 23 2025 /usr/lib/telnetlogin
root@localhost:/bin# chown root:root /usr/lib/telnetlogin
root@localhost:/bin# ls /usr/lib/telnetlogin -lh
-rwxr-xr-x 1 root root 11K Jul 23 2025 /usr/lib/telnetlogin
root@localhost:/bin# chmod 4755 /usr/lib/telnetlogin
root@localhost:/bin# ls /usr/lib/telnetlogin -lh
-rwsr-xr-x 1 root root 11K Jul 23 2025 /usr/lib/telnetlogin
root@localhost:/bin#
```

6.4 telnetd 登录很慢

能够登录，但是要等 5~10 s 才能登录。/etc/systemd/network/20-static-eth0.network 文件里不要配置 DNS，若需配置 DNS 在 /etc/resolv.conf 里配置

```
root@localhost:~# cat /etc/systemd/network/20-static-eth0.network
[Match]
Name=eth0

[Network]
Address=192.168.5.100/24
Gateway=192.168.5.1
DNS=114.114.114.114
root@localhost:~#
```

6.5 海鸥派预装软件

```
apt install iperf3
apt install udhcpd
apt install wpasupplicant
apt install hostapd
apt install tftp
apt install tftp-hpa
apt install ethtool
apt install iproute2
apt install can-utils

apt install dbus
apt install bluez
apt install bluetooth

systemctl stop udhcpd           # 停止当前运行
systemctl disable udhcpd       # 禁用开机自启
systemctl mask udhcpd          # 防止任何启动（彻底屏蔽）
systemctl status udhcpd

systemctl stop hostapd
systemctl disable hostapd
systemctl mask hostapd
systemctl status hostapd

systemctl stop wpa_supplicant
systemctl disable wpa_supplicant
systemctl mask wpa_supplicant
systemctl status wpa_supplicant

systemctl enable serial-getty@ttyAMA0.service
systemctl start serial-getty@ttyAMA0.service

sudo chroot Ubuntu22.04 apt clean
sudo chroot Ubuntu22.04 apt autoremove

apt-get install psmisc          #killall 命令

apt-get install libgomp1        #yolo 运行需要的库， 暂未预装
```



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